

EMMANUEL

Iteration 1

Given Data.

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \text{Learning rate} = 0.001$$

$e = \text{error.}$

$$\hat{y} = y_{\text{pred}} = m_1 x_1 + m_2 x_2 + b = X \cdot m + b.$$

Prediction

$$\begin{aligned} X \cdot m &= (1)(-1) + (3)(2) = -1 + 6 = 5 \\ (4)(-1) + (10)(2) &= -4 + 20 = 16 \end{aligned} \Rightarrow y_{\text{pred}} = \begin{bmatrix} 5 + 1 \\ 16 + 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

MSE

$$e = y_{\text{pred}} - y = \begin{bmatrix} 6 - 5 \\ 17 - 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$MSE = \frac{1}{n} \sum_i (y_{\text{pred}} - y)^2 = \frac{1}{2} (1^2 + 11^2) = \frac{1}{2} (1 + 121) = 61$$

Chain Rule

with respect to m :

$$\frac{\partial J}{\partial m_j} = \frac{\partial J}{\partial y_{\text{pred}}} \cdot \frac{\partial y_{\text{pred}}}{\partial m_j} = \frac{2}{n} \sum_i (y_{\text{pred}} - y) x_{ji} = \frac{\partial J}{\partial m} = \frac{2}{n} X^T e$$

with respect to b (scalar)

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial y_{\text{pred}}} \cdot \frac{\partial y_{\text{pred}}}{\partial b} = \frac{2}{n} \sum_i (y_{\text{pred}} - y) = \frac{\partial J}{\partial b} = \frac{2}{n} \sum_i e$$

Plug in numbers ($n=2$ so $\frac{2}{n}=1$).

$$\frac{\partial J}{\partial m} = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$m_1 = (1)(1) + (4)(11) = 45$$

$$m_2 = (3)(1) + (10)(11) = 113$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}, \frac{\partial J}{\partial b} = \sum e_i = 1 + 11 = 12$$

Parameter - Learning rate $\cdot$ gradient.

$$\text{new } m_1 = -1 - 0.001(45) = -1 - 0.045 = -1.045$$

$$\text{new } m_2 = 2 - 0.001(113) = 2 - 0.113 = 1.887$$

$$\text{new } b = 1 - 0.001(12) = 1 - 0.012 = 0.988$$

Iteration 2

$$m = [-1.045, 1.887] , b = 0.988$$

~~Prediction~~: $\hat{y} = X_m + b$

$$X_m = (1)(-1.045) + 3(1.887) = -1.045 + 5.661 = 4.616$$

$$= 4(-1.045) + 10(1.887) = -4.180 + 18.870 = 14.690$$

$$\hat{y} = X_m + b :$$

$$\Rightarrow 4.616 + 0.988 = 5.604$$

$$\Rightarrow 14.690 + 0.988 = 15.678$$

$$\hat{y} = [5.604, 15.678]$$

for MSE:

$$e = \hat{y} - y = [5.604 - 5, 15.678 - 6] = [0.604, 9.678]$$

$$MSE = \frac{1}{2} ((0.604)^2 + (9.678)^2) = \frac{1}{2} (0.365 + 93.664)$$

$$= 47.01$$

Gradients

$$\frac{\partial J}{\partial m} = X^T e ,$$

$$m_1 = (1)(0.604) + 4(9.678) = 39.316$$

$$m_2 = 3(0.604) + 10(9.678) = 98.592$$

$$\frac{\partial J}{\partial b} = \sum e = 0.604 + 9.678 = 10.282$$

New values

$$new = old - (0.001 \times gradient) :$$

$$m_1 = -1.045 - 0.001(39.316) = -1.084$$

$$m_2 = 1.887 - 0.001(98.592) = 1.788$$

$$b = 0.988 - 0.001(10.282) = 0.978$$

Iteration 3.

$$m = [-1.084, 1.788], \quad b = 0.978$$

$$\text{Prediction : } \hat{y} = X_m + b.$$

$$X_m = (1)(-1.084) + (3)(1.788) = -1.084 + 5.364 = 4.280.$$

$$= (4)(-1.084) + (10)(1.788) = -4.336 + 17.880 = 13.544,$$

$$\hat{y} = X_m + b$$

$$\Rightarrow 4.280 + 0.978 = 5.258.$$

$$\Rightarrow 13.544 + 0.978 = 14.522$$

$$\hat{y} = [5.258, 14.522]$$

For MSE.

$$e = \hat{y} - y = [5.258 - 5, 14.522 - 6] = [0.258, 8.522].$$

$$MSE = \frac{1}{2} (0.258^2 + 8.522^2) = \frac{1}{2} (0.067 + 72.625)$$

$$= \frac{1}{2} (72.692) = 36.35.$$

Gradients.

$$\frac{\partial J}{\partial m} = X_e^T.$$

$$m_1 = (1)(0.258) + (4)(8.522) = 0.258 + 34.088 = 34.346$$

$$m_2 = (3)(0.258) + (10)(8.522) = 0.774 + 85.220 = 85.994.$$

$$\frac{\partial J}{\partial b} = \sum e = 0.258 + 8.522 = 8.780.$$

New values

$$new = old - 0.001 \times \text{gradient}$$

$$m_1 = -1.084 - 0.001(34.346) = -1.118$$

$$m_2 = 1.782 - 0.001(85.994) = 1.702$$

$$b = 0.998 - 0.001(8.780) = 0.969$$

$$y_{est} = 1.118x + 0.969 = (1.118)(0) + (0.969)(1) = 0.969$$

$$y_{est} = 1.118x + 0.969 = (1.118)(10) + (0.969)(1) = 11.279$$

$$y = mx + b$$

$$-922.0 = 949.0 + 0.969 \times 0 =$$

$$-922.0 = 949.0 + 0.969 \times 0 =$$

$$[-922.0, 949.0] = 0$$

error

$$[922.0, 949.0] = [0 - 922.0, 0 - 949.0] = y - \hat{y} = 0$$

$$(922.0 + 949.0) \times \frac{1}{2} = (922.0 + 949.0) \times \frac{1}{2} = 935.5$$

$$935.5 = (949.0) \times \frac{1}{2} =$$

gradient

$$\frac{\partial}{\partial m} X = \frac{\partial}{\partial m}$$

$$922.0 = 949.0 + 0.969 = (1.118)(0) + (0.969)(1) = 0.969$$

$$949.0 = 922.0 + 0.969 = (1.118)(10) + (0.969)(1) = 11.279$$

$$-922.0 = 949.0 + 0.969 = 0.969 = \frac{0}{0.969}$$

4th Iteration

$$m = \begin{bmatrix} -1.118 \\ 1.702 \end{bmatrix}, \quad b = \begin{bmatrix} 0.969 \\ 0.969 \end{bmatrix}, \quad y = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\hat{y} = X m + b$$

$$X m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.118 \\ 1.702 \end{bmatrix} \\ = \begin{bmatrix} -1.118 + (3 \times 1.702) \\ (4 \times -1.118) + (10 \times 1.702) \end{bmatrix} = \begin{bmatrix} 3.988 \\ 12.548 \end{bmatrix}$$

$$\hat{y} = X m + b, \quad \begin{bmatrix} 3.988 \\ 12.548 \end{bmatrix} + \begin{bmatrix} 0.969 \\ 0.969 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 4.957 \\ 13.517 \end{bmatrix}$$

$$e = \hat{y} - y, \quad \begin{bmatrix} 4.957 \\ 13.517 \end{bmatrix} - \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} -0.043 \\ 7.517 \end{bmatrix}$$

$$MSE(J) = \frac{1}{n} \sum (\hat{y} - y)^2 \\ = \frac{1}{2} \sum (-0.043, 7.517)^2 \\ = \frac{1}{2} [(-0.043)^2 + (7.517)^2] \\ = \underline{\underline{28.25369}}$$

Gradient

$$\frac{\partial J}{\partial m} = X^T e$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -0.043 \\ 7.517 \end{bmatrix} = \begin{bmatrix} -0.043 + (4 \times 7.517) \\ (3 \times -0.043) + (10 \times 7.517) \end{bmatrix} \\ = \begin{bmatrix} 30.025 \\ 75.041 \end{bmatrix}$$

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m} \\ = \begin{bmatrix} -1.118 \\ 1.702 \end{bmatrix} - 0.001 \begin{bmatrix} 30.025 \\ 75.041 \end{bmatrix} \\ = \begin{bmatrix} -1.118 \\ 1.702 \end{bmatrix} - \begin{bmatrix} 0.030025 \\ 0.075041 \end{bmatrix} \\ m_{new} = \begin{bmatrix} -1.148025 \\ 1.626959 \end{bmatrix} //$$

$$\frac{\partial J}{\partial b} = \sum c$$

$$= -0.043 + 7.517$$

$$= 7.474$$

$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b}$$

$$b_{new} = 0.963 - 0.001(7.474)$$

$$b_{new} = \underline{\underline{0.9615}}$$

Therefore, $m_{new} = \begin{bmatrix} -1.148025 \\ 1.626959 \end{bmatrix}$ $b_{new} = \begin{bmatrix} 0.9615 \\ 0.9615 \end{bmatrix}$